

Title

Nutritional outcomes and gastrointestinal tolerance following a systematic ERAS-derived feeding protocol in ICU patients

Abstract

Background: Implementation of systematic, recovery-focused nutritional protocols in intensive care may improve nutritional adequacy while reducing complications. **Methods:** 100 critically ill patients were randomly assigned to receive either a structured ERAS-derived nutrition protocol or standard ICU nutritional support. Outcomes including daily macronutrient intake, hemoglobin and serum albumin levels, rates of gastrointestinal adverse events, and days spent in intensive care were measured. **Results:** The protocol-guided intervention group achieved significantly higher mean daily caloric intake [1089 – 246 kJ/d vs 891 – 198 kJ/d; $p < 0.05$] and protein delivery [105.2 – 14.1 g/d vs 98.4 – 11.8 g/d; $p < 0.05$]. Hemoglobin (119.3 – 7.2 g/L vs 108.1 – 6.8 g/L) and albumin levels (46.1 – 3.4 g/L vs 42.3 – 3.9 g/L) were significantly higher in the protocol group ($p < 0.05$). Gastrointestinal complications were reduced in the intervention group: retention 19.0% vs 39.0%, diarrhea 11.0% vs 23.0%, abdominal discomfort 13.0% vs 31.0%, and bleeding 5.0% vs 11.0%. Intensive care duration was shortened in the protocol group [9.4 – 3.6 days vs 12.1 – 4.2 days; $p < 0.05$]. **Conclusions:** Structured ERAS-informed nutrition protocols improve nutrient delivery and gastrointestinal outcomes in critically ill patients.